

Foundations of Olympiad Analytics

GRADE 3 (AGES 7-9)

BECOME AN OLYMPIAD DETECTIVE! 

YOUR MISSION BRIEFING

Welcome, Young Detective! Olympiad questions are like secret puzzles. They don't just ask you to do simple school math or repeat science words. Instead, they give you **hidden patterns** and **mysteries** to solve. Today, you will learn the top-secret tricks to crack these codes like a true grandmaster!



Part 1: The Math Detective (Pattern Decryption)

In school, you might see easy patterns like **2, 4, 6, 8**. But Olympiad puzzles are **multi-variable**. That means more than one thing changes at the exact same time!



Case Study: The Shifting Triangles

Imagine a pattern sequence where shapes change their position and number of dots together:

- **Term 1:** A standard upright triangle pointing **North** with **1** dot inside.
- **Term 2:** The triangle twists **90°** Clockwise (pointing **East**) and now has **3** dots inside.
- **Term 3:** It twists another **90°** Clockwise (pointing **South**) and has **5** dots inside.

How to crack it? Don't look at everything at once! Split the puzzle into two pieces:

1. **The Twist (Orientation):** It spins **90°** every step. Because a full circle has 4 turns, the shape will face the exact same way every 4 steps! So, **Term 5** will look exactly like **Term 1** (pointing North).
2. **The Dots (Arithmetic Growth):** The dots go up by **+2** each time: **1, 3, 5, 7, 9...** (Odd numbers!). For **Term 5**, it will have **9** dots.



Real-World Anchor: The Magic of Kolam & Rangoli

Have you seen beautiful traditional Indian Kolam or Rangoli grid designs drawn with lines around dots? These use the exact same math rules! A small 4x4 dot grid can expand perfectly into a giant 8x8 design just by following a repeating symmetrical rule. Math is all around us in our festivals!



Part 2: The Science Detective (Molecular Clues)

Let's look past what things look like on the outside and peek at their tiny invisible building blocks called **molecules**!



1. Solid Phase (The High-Density Lock)

Inside a solid item, like an iron **Tawa** (griddle) or a clay **Diya**, the tiny particles are locked tightly in a strong grid. They can't move around; they only vibrate in place. This is why a Tawa never changes its shape when you cook on it!



2. Liquid Phase (The Sliding Clusters)

In liquids, the particles have a little bit of space to slide, roll, and flow past each other. The total amount stays the same, but it changes shape to fit whatever it's inside!

The Ghee Experiment: Think about pure **Ghee** or **Coconut Oil**. In the cold winter months, the particles lose energy and lock into a hard, white solid. But when summer comes, they absorb heat, break their rigid locks, and turn into a clear liquid that fits perfectly into a steel **Katori** (bowl) or glass jar!



3. Gas Phase (The Free Sprinters)

Gas particles have absolutely zero bonds holding them. They fly around super fast in every direction! They have no fixed shape and no fixed volume.

The Agarbatti Mystery: When an **Agarbatti** (incense stick) is lit in a quiet Pooja corner, its sweet-smelling aromatic gas molecules fly out and quickly travel across the entire house. This fast spreading is called **gas diffusion**!



Part 3: The Logic Detective (Arrangements & Ladders)

Olympiad logic questions try to fill your head with too many clues at once. Never try to guess or hold them all in your brain! Always draw a **Height Ladder** or a seating row line on paper.



Case Study: The Bengaluru Bus Queue

Let's map five friends waiting for their school bus based on these clues:

- Clue 1: Arjun is taller than Charan, but shorter than Diya. (*Diya > Arjun > Charan*)
- Clue 2: Bhavana is shorter than Charan. (*Charan > Bhavana*)
- Clue 3: Diya is shorter than Esha. (*Esha > Diya*)

By connecting these vertical pieces like steps on a ladder, we get the absolute truth instantly:

Esha (Tallest) > Diya > Arjun > Charan > Bhavana (Shortest)

We do the exact same thing when mapping out family members sitting in a row for a traditional festive floor **Pangat** meal!

Part 4: The 30-Second "Olympiad Hack" Sprint

When the timer is ticking down during a real test, you don't always need to solve the whole question to find the answer. Use the **Process of Elimination** to throw away trap options!

- **Step 1:** Find a single rule (e.g., "The answer must be an even number" or "The shape must point East").
- **Step 2:** Look at the choices and cross out every single option that breaks your rule. This instantly removes the "distractor" traps!

Part 5: The Olympiad Challenge Arena

Put on your detective hat! Let's solve these handpicked challenge questions using your new super-skills.

Q1 • MATH The Coin Step Puzzle

An arithmetic coin pattern grows at each step. Step 1 has three ₹2 coins. Step 2 has five ₹2 coins. Step 3 has seven ₹2 coins. What is the total **monetary value** (money amount) of the coins at Step 6?

A) 13 coins	B) ₹22
C) ₹26	D) ₹11

 **Detective Solution:** The number of coins follows an odd number rule: **3, 5, 7, 9, 11, 13...** At Step 6, there are **13** coins. But watch out for the trap! The question asks for the money value, not the number of coins. **13 times ₹2 = ₹26. Correct Option: C**

Q2 • SCIENCE The Mystery Container Experiment

Radhika squeezes three sealed containers holding unknown substances X, Y, and Z. Container X won't shrink at all. Container Y gets squeezed down by over half instantly. Container Z keeps its exact volume but changes its shape to match the container it's poured into. What is substance Y?

A) An iron block	B) Compressed air smoke
C) Liquid coconut oil	D) A solid wood brick

 **Detective Solution:** High compressibility (being squeezed easily) is the special superpower of a **gas** because its molecules have tons of free space! Since Y shrinks easily, it must be a gas. Compressed air smoke is the only gas option here. **Correct Option: B**

Q3 • LOGIC The Delhi Lunch Line

In a school lunch line, Aarav is standing directly behind Kabir. Kabir is standing three places ahead of Meera. If Aarav is the 4th person from the very front of the line, what is Meera's exact numerical placement from the front?

A) 5th	B) 6th
C) 7th	D) 3rd

 **Detective Solution:** Let's draw the line positions: If Aarav is 4th, and he is directly behind Kabir, then Kabir must be 3rd. Since Kabir is 3 places ahead of Meera, we add 3 to Kabir's spot: $3 + 3 = 6$. So Meera is 6th!

Correct Option: B

Q4 • MATH The Combined Climate Code (Grandmaster Challenge)

An automated climate chamber changes its state based on a math pattern code. It starts at Step 1 with a solid block of ice at -10°C . Every next step raises the temperature by exactly 4°C . Water turns to liquid at 0°C and turns to vapor gas at 100°C . What is the state and temperature at Step 30?

A) Solid Ice at 110°C	B) Liquid Water at 106°C
C) Water Vapor Gas at 106°C	D) Liquid Water at 120°C

 **Detective Solution:** From Step 1 to Step 30, there are 29 increases. Total temperature increase = $29 \text{ times } 4^{\circ}\text{C} = 116^{\circ}\text{C}$. The starting temperature was -10°C , so $-10 + 116 = 106^{\circ}\text{C}$. Since 106°C is hotter than the boiling point (100°C), the water has boiled away into a gas! **Correct Option: C**

Q5 • LOGIC The Festival Seating Mystery

Four family members—Grandfather, Father, Deepa, and Rahul—are sitting in a straight line on floor mats. Deepa sits immediately next to Grandfather. Father sits next to Deepa, but is NOT next to Grandfather. If Rahul is at position 1 on the absolute left edge, who is sitting at position 4 on the absolute right edge?

A) Grandfather	B) Father
C) Deepa	D) Information insufficient

 **Detective Solution:** Since Deepa sits next to both Grandfather and Father, she must be directly in the middle between them (Grandfather - Deepa - Father). Since Rahul is at spot 1, the remaining three spots must be filled by this group. To keep Father away from Grandfather, the order must be: 1: Rahul, 2: Grandfather, 3: Deepa, 4: Father. Therefore, Father is at spot 4! **Correct Option: B**